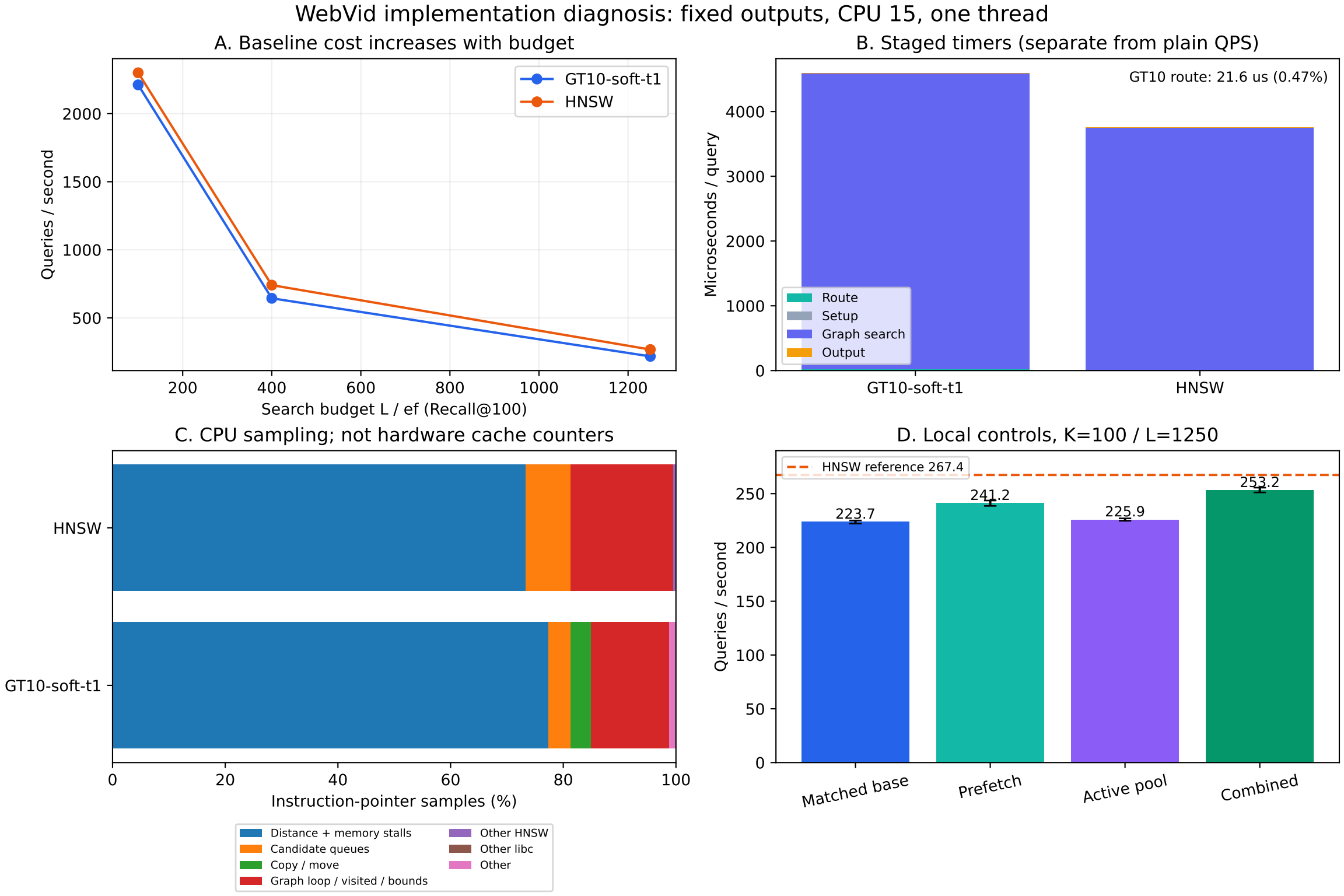
WebVid implementation diagnosis: fixed outputs, CPU 15, one thread
A. Baseline cost increases with budget
GT10-soft-t1
HNSW
Queries / second
2000
1500
1000
500
200
400
600
800
1000
1200
Search budget L / ef (Recall@100)
B. Staged timers (separate from plain QPS)
GT10 route: 21.6 us (0.47%)
Microseconds / query
4000
3000
2000
1000
0
Route
Setup
Graph search
Output
GT10-soft-t1
HNSW
C. CPU sampling; not hardware cache counters
HNSW
GT10-soft-t1
0
20
40
60
80
100
Instruction-pointer samples (%)
Distance + memory stalls
Candidate queues
Copy / move
Graph loop / visited / bounds
Other HNSW
Other libc
Other
D. Local controls, K=100 / L=1250
HNSW reference 267.4
Queries / second
250
200
150
100
50
0
223.7
241.2
225.9
253.2
Matched base
Prefetch
Active pool
Combined